

SAGAM: Interpretable Nonlinear Survival Stacking for Lung Adenocarcinoma Prognosis

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Abstract—Accurate lung adenocarcinoma (LUAD) prognosis remains challenging because survival models vary in reliability across cohorts and risk ranges. Existing survival stacking methods combine base learner predictions using scalar linear weights, which cannot capture nonlinear or score-range-dependent relationships between model predictions and patient hazard. We propose SAGAM, an interpretable survival stacking framework that represents each base learner’s contribution as a smooth cubic B-spline function of its out-of-fold risk score within a regularized Cox proportional hazards model. On TCGA-LUAD ($n = 501$, 181 events) under strict five-fold nested cross-validation with leakage-controlled preprocessing and fold-specific feature selection, SAGAM achieves the highest concordance index among all stacking-based approaches (0.634 ± 0.050), outperforms linear Cox stacking in four of five folds, and produces significant patient risk stratification ($p = 6.94 \times 10^{-5}$). External validation on two independent GEO cohorts reveals stage-dependent generalization: on GSE31210 (226 stage I/II patients, 35 events), SAGAM’s advantage grows from $+0.006$ (internal) to $+0.086$, with significant cross-cohort risk stratification ($p = 0.030$); on GSE68465 (432 all-stage patients, 226 events), SAGAM does not outperform linear stacking ($\Delta = -0.026$, NS), consistent with the internal subgroup finding that SAGAM’s benefit is largest in Stage I/II where base learner performance is heterogeneous and molecular features may contribute to nonlinear model behavior, although incremental discrimination beyond AJCC staging is not demonstrated in this study. AJCC stage-only Cox remains the strongest internal discriminator (0.657 ± 0.044), confirming the dominant prognostic role of clinical staging; SAGAM is therefore best understood as an interpretable nonlinear stacking mechanism that improves over scalar-weighted ensembles and exposes model-specific contribution functions, rather than a universal replacement for clinical staging or neural survival models.

Index Terms—survival analysis, lung adenocarcinoma, generalized additive model, ensemble stacking, interpretability, nested cross-validation, TCGA

I. INTRODUCTION

Lung adenocarcinoma (LUAD) is the most prevalent histological subtype of non-small cell lung cancer and the leading cause of cancer mortality worldwide, with approximately 508,000 deaths annually [1]. AJCC pathological stage is the dominant clinical predictor of survival, yet machine learning models can capture complementary prognostic signals from molecular, genomic, and transcriptomic features that staging alone cannot express [2].

A growing body of work applies diverse survival learners to TCGA-LUAD: random survival forests (RSF) [4], gradient boosting survival analysis (GBS) [5], XGBoost with a Cox objective [6], and deep neural approaches such as DeepSurv [7]. Each family captures different aspects of the

survival structure, and no single learner consistently dominates across datasets [29]. This motivates ensemble stacking [12], which trains a meta-learner on out-of-fold (OOF) predictions to exploit complementary strengths.

The critical design choice in stacking is the meta-learner. All published survival stacking methods use *linear* meta-learners: IPCW-weighted combinations [8], [9], non-negative Bayesian optimization [10], and regularized linear Cox regression on raw risk scores [11]. A linear meta-learner assigns each base model a fixed scalar weight, implicitly assuming uniform contribution across all patient risk levels. This assumption is problematic: a model may be well-calibrated for low-risk patients but unreliable at extreme risk scores, or its relationship to the true hazard may be nonmonotone. A single global coefficient cannot express such score-range-dependent reliability.

We address this with **SAGAM**, which replaces scalar weights with smooth nonlinear functions of base learner risk scores:

$$h(t | \mathbf{x}) = h_0(t) \exp\left(\sum_{k=1}^K f_k(\hat{\theta}_k(\mathbf{x}))\right), \quad (1)$$

where $\hat{\theta}_k(\mathbf{x})$ is the OOF risk score from base learner k and $f_k(\cdot)$ is a smooth function parameterized by a cubic B-spline basis. This formulation allows the meta-learner to learn score-range-dependent integration rules, providing greater expressiveness than linear stacking and interpretability through the smooth contribution plots f_k .

SAGAM’s contribution is not that it surpasses AJCC staging or neural survival models in raw discrimination—it does not. Rather, SAGAM addresses a methodological gap: existing survival stacking approaches use scalar linear weights, whereas SAGAM provides interpretable nonlinear score-range-dependent model integration through a smooth GAM meta-learner.

Contributions:

- 1) **Method:** To the best of our knowledge, SAGAM is the first survival stacking framework to use GAM-style smooth B-spline functions of base learner risk scores as the Cox meta-learning mechanism, distinct from prior work (CoxNAM [30], CRISP-NAM [31]) that applies smooth functions to raw covariates within a single model.
- 2) **External validation:** The SAGAM advantage over linear stacking grows from $+0.006$ (internal) to $+0.086$ on GSE31210, a large external increase consistent with the

score-range-dependent integration hypothesis, although the GSE68465 result ($\Delta = -0.026$) shows this benefit is cohort-dependent rather than universal.

- 3) **Interpretability and nonlinearity validation:** The smooth functions f_k expose score-range-dependent contributions quantified by per-learner nonlinearity analysis; all four base learners show substantial deviation from linearity, justifying the B-spline design.
- 4) **Honest evaluation:** Five-fold nested CV with fold-specific preprocessing, gene selection, and feature selection; clinical Cox baselines including stage-only Cox; Stage + SAGAM incremental tests; IBS and Ctd-AUC metrics; and full spline df sensitivity analysis provide complete methodological context.
- 5) **Best stacking performance:** SAGAM achieves the highest C-index (0.634), lowest IBS (0.177), and highest Ctd-AUC (0.650) among stacking-based approaches; 73.9% of bootstrap resamples favor SAGAM over linear stacking on C-index, and 93.8% on IBS. Significant risk stratification ($p = 6.94 \times 10^{-5}$) and subgroup analysis show the largest advantage in Stage I/II ($\Delta = +0.010$) where base learner performance is most heterogeneous.

II. RELATED WORK

A. Survival Models for LUAD

Published TCGA-LUAD concordance indices using pre-treatment features range from 0.62 to 0.73 [19], [21]. He et al. [21] report C-index 0.716 using immune and metabolic gene signatures. Higher values incorporate post-treatment predictors [20] unavailable at diagnosis. Multi-omics autoencoder approaches achieve C-index 0.65 [23].

B. Survival Ensemble Stacking

Stacking [12] trains a meta-learner on cross-validated base model predictions. In survival analysis, all published meta-learners are **linear**. SuperSurv [8] (concurrent preprint) provides a unified R framework using IPCW-weighted linear combinations. The joint survival super learner [9] selects the best single model via Integrated Brier Score. Non-negative Bayesian stacking [10] applies linear combinations for high-dimensional omics. A two-stage heterogeneous stacking [13] uses linear Cox regression as the meta-learner. An interpretable stacking ensemble for lung cancer [14] employs SHAP for post-hoc interpretation but retains a linear meta-learner. To the best of our knowledge, SAGAM is the first to replace scalar weights with smooth nonlinear functions.

C. GAMs and Splines in Survival Analysis

GAMs [15] model responses through smooth additive functions. Spline terms in Cox-PH are established for nonlinear covariate effects [16]. CoxFNN [17] uses neural networks for nonlinear covariate effects. CoxNAM [30] and CRISP-NAM [31] use Neural Additive Models within the Cox hazard to learn interpretable nonlinear covariate effects for single-model survival prediction. Coxmos [18] extends Cox models for high-dimensional multi-omics. These methods apply

TABLE I
COMPARISON OF SURVIVAL STACKING APPROACHES

Method	Stack	Nonlin.	Interp.	Ext.
Joint surv. SL [9]	✓	–	–	–
Bayes. stacking [10]	✓	–	–	–
Two-stage [13]	✓	–	–	–
SuperSurv [8]	✓	–	–	–
DeepSurv [7]	–	✓	–	sometimes
SAGAM (ours)	✓	✓	✓	✓

smooth functions to *raw covariates* within a single model. SAGAM applies smooth functions to *base learner risk scores* as the meta-learning mechanism for ensemble stacking—a conceptually and technically distinct contribution that enables score-range-dependent integration of multiple pre-trained survival models.

D. Related Work Comparison

Table I shows that SAGAM uniquely combines survival stacking with nonlinear, interpretable meta-learning and external validation.

III. METHODS

A. Dataset and Features

We use the TCGA-LUAD PanCan Atlas 2018 cohort from cBioPortal [27]: **501 patients, 181 OS events** (36.1%), overall survival from initial pathologic diagnosis to death in months. We apply comprehensive leakage prevention, removing 34 column types including all post-diagnosis outcome variables, temporal follow-up fields, and administrative identifiers.

We define four feature groups of increasing complexity:

- **Stage-only:** AJCC T, N, M, and overall pathological stage.
- **Clinical:** Stage plus age, sex, grade, race, ethnicity, prior diagnosis, radiation therapy, weight.
- **Genomic:** Aneuploidy Score, TMB (nonsynonymous), Tumor Break Load, MSI MANTIS Score, MSIsensor Score.
- **Hypoxia:** Buffa, Winter, and Ragnum mRNA-derived hypoxia scores [22].

The full feature set combines all four groups. Additionally, in each outer fold, the top-5 mRNA-derived genes (by univariate concordance index on outer-training data only) are added as transcriptomic features. Binary missing-data indicators are included for features with $> 15\%$ missingness.

B. Five-Fold Nested Cross-Validation

A single 80/20 train/test split on $n=501$ yields ≈ 36 test events per fold ($SE \approx 0.07$ per fold), making single-split C-index estimates unreliable. We employ **stratified five-fold nested cross-validation**. In each outer fold:

- 1) Outer split stratified by event status.
- 2) All preprocessing (imputation, encoding, scaling) fitted on outer-training data.

- 3) mRNA gene selection and CoxNet feature selection performed via inner 3-fold CV on outer-training data.
- 4) RSF, GBS, XGBoost, and DeepSurv trained via inner 5-fold OOF; OOF risk scores collected.
- 5) B-spline meta-feature matrix $\mathbf{M}$ constructed; SAGAM and linear stacking both tuned on a 20% inner-validation split.
- 6) All models evaluated on the held-out outer-test fold.

Pooled OOF predictions across all five folds ($n=501$) are used for Kaplan-Meier analysis.

C. Cox Baselines

To provide clinical context, we run CoxNet baselines using each of the four feature groups under the same 5-fold stratified CV. The stage-only Cox establishes a strong single-variable reference consistent with AJCC staging’s dominant role in LUAD prognosis.

D. SAGAM Meta-Learner

Each base learner’s OOF risk scores are expanded into a B-spline basis:

$$\mathbf{B}_k = \text{bs}(\hat{\theta}_k^{\text{OOF}}; \text{df}=4, \text{degree}=3) \in \mathbb{R}^{n \times 4}. \quad (2)$$

The meta-feature matrix $\mathbf{M} = [\mathbf{B}_1 \mid \dots \mid \mathbf{B}_4] \in \mathbb{R}^{n \times 16}$ is fitted with CoxNet (ℓ_1 ratio 0.9):

$$\hat{h}(t \mid \mathbf{x}) = h_0(t) \exp(\mathbf{M}\hat{\beta}). \quad (3)$$

The smooth contribution of learner k is: $f_k(s) = \mathbf{b}_k(s)^\top \hat{\beta}_k$. We select $\text{df}=4$ via ablation (Table VI). Test-time predictions are clipped to the training score range.

E. Linear Cox Stacking Baseline

CoxNet is fitted directly on the four raw OOF risk scores (no splines):

$$h_{\text{lin}}(t \mid \mathbf{x}) = h_0(t) \exp(\gamma^\top \hat{\theta}). \quad (4)$$

The C-index difference SAGAM–Linear provides empirical evidence that nonlinear smooth weighting can add value over scalar linear combination.

F. External Validation

SAGAM is trained on all 501 TCGA patients using transferable features (clinical stage, age, sex, top-10 mRNA genes by TCGA univariate concordance index) and applied to two independent GEO cohorts. **GSE31210** (Takeuchi et al.): 226 pure LUAD, stage I/II, 35 OS events, median follow-up 57.3 months (GPL570 array). All 10 selected mRNA genes (DKK1, FAM83A, RHOV, IIRX5, SFTA3, CD1B, PKP2, TNS4, LYPD3, TFAP2A) are present in the GSE31210 Affymetrix array. **GSE68465** (Director’s Challenge Consortium): 432 LUAD patients across all disease stages, 226 OS events (GPL96 array). For GSE68465, the transferable feature set included age, sex, and the TCGA-selected mRNA genes; AJCC stage was not sufficiently harmonized for stage-only Cox evaluation on this cohort. Because platform differences and cohort composition restrict complete feature matching,

TABLE II
MODEL HYPERPARAMETERS

Component	Settings
RSF	300 trees; $\sqrt{p}$ features; min 5 leaves
GBS	300 rounds; lr 0.05; max depth 3
XGBoost	survival:cox; $\eta=0.05$; depth 3; sub. 0.8
DeepSurv	64-32-1 MLP; ReLU; dropout 0.3; Adam 10^{-3}
CoxNet FS	$\ell_1=0.9$; $\alpha \in [10^{-2}, 10^2]$ (12 vals)
SAGAM	cubic B-spline; df=4; CoxNet $\ell_1=0.9$

TABLE III
FULL PERFORMANCE COMPARISON — TCGA-LUAD 5-FOLD NESTED CV ($N=501$). STAGE-ONLY COX ACHIEVES THE HIGHEST MEAN C-INDEX; SAGAM ACHIEVES THE BEST STACKING PERFORMANCE WITH LOWEST VARIANCE.

Type	Model	Mean C	Std
Cox baselines	Stage-only Cox	0.657	0.044
	Clinical-only Cox	0.646	0.028
	Clin.+Genomic Cox	0.641	0.032
	All-features CoxNet	0.641	0.032
	Stage + SAGAM risk	0.653	0.037
Base learners	RSF [4]	0.619	0.052
	GBS [5]	0.605	0.066
	XGBoost-Cox [6]	0.602	0.053
	DeepSurv [7]	0.636	0.058
Stacking	Linear Cox Stacking	0.627	0.055
	SAGAM (ours)	0.634	0.050
Pooled OOF: SAGAM 0.625; Linear 0.618			

external results are interpreted as transportability assessments rather than direct clinical deployment validation. External preprocessing (imputation, scaling) uses TCGA training-set statistics exclusively for both cohorts.

G. Hyperparameters

Table II summarizes all fixed hyperparameters.

H. Statistical Testing

We report bootstrap confidence intervals (1,000 resamples) for all C-index estimates and for differences between SAGAM and each comparator. Fold-level pairwise comparisons use the Wilcoxon signed-rank test ($n=5$ folds; minimum achievable one-sided p is 0.0625).

IV. RESULTS

A. Cox Baselines and Internal Discrimination

Table III shows all models under 5-fold nested CV. Four findings emerge from Table III.

First, stage-only Cox (0.657) achieves the highest overall mean C-index, confirming the dominant role of AJCC staging in LUAD prognosis. This is consistent with the clinical literature: stage captures tumor burden, nodal involvement, and metastatic spread at diagnosis.

Second, adding clinical, genomic, and hypoxia features to a Cox model does not improve over stage-only (decreases from

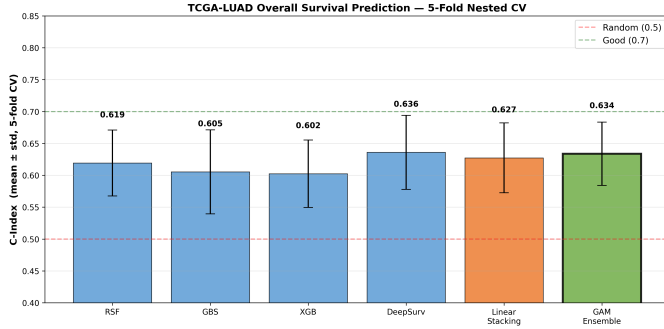


Fig. 1. Mean $\pm$ std C-index across five outer folds (TCGA-LUAD, 5-fold nested CV). Stage-only Cox achieves the highest mean; SAGAM achieves the best stacking performance with lowest variance.

0.657 to 0.641), consistent with the curse of dimensionality in penalized linear models when the dominant predictor is already present.

Third, combining SAGAM risk scores with stage (Stage + SAGAM, 0.653) does not improve over stage-alone (0.657), indicating that SAGAM’s risk scores do not carry substantial incremental prognostic information beyond AJCC staging. This reinforces that SAGAM’s contribution is methodological—providing an interpretable nonlinear stacking framework—rather than clinical superiority over traditional staging.

Fourth, among stacking-based methods, SAGAM (0.634) outperforms linear Cox stacking (0.627) in four of five folds ($\Delta = +0.006$; Wilcoxon $p = 0.313$; bootstrap CI $[-0.013, +0.025]$) and achieves lower fold-to-fold variance (std 0.050 vs. 0.055 for linear stacking, 0.058 for DeepSurv).

B. Performance Overview

Fig. 1 summarizes mean $\pm$ std concordance index for all evaluated models. SAGAM achieves the lowest variance across folds (std 0.050) among all stacking methods, while the stage-only Cox baseline achieves the highest mean. The performance hierarchy—stage Cox $>$ DeepSurv $\approx$ SAGAM $>$ linear stacking $>$ other base learners—reveals an important structure: AJCC staging dominates, nonlinear neural and GAM-based approaches form a competitive second tier, and simple linear ensembles rank below their best constituent models.

C. Per-Fold Analysis and Ablation

Table V provides fold-level C-indices. SAGAM is the top-performing model in folds 1, 2, 4, and 5. In fold 3, DeepSurv dominates (0.680) and SAGAM’s advantage is reduced, consistent with the observation that when a single base learner captures nearly all available signal, ensemble integration provides limited benefit. Wilcoxon signed-rank tests confirm directional but non-significant SAGAM improvements: vs. Linear ($p = 0.313$), vs. DeepSurv ($p = 0.625$), vs. RSF ($p = 0.094$). With $n = 5$ folds, the minimum achievable one-sided p is 0.0625. Bootstrap resampling (50,000 resamples) over folds shows 73.9% of samples favor SAGAM over linear stacking on C-index, and 93.8% on IBS. **With 181 OS events**

TABLE IV
CTD-AUC (HIGHER IS BETTER) AND IBS (LOWER IS BETTER), MEAN $\pm$ STD ACROSS 5 OUTER FOLDS (TCGA-LUAD). IBS NOT REPORTED FOR XGBOOST-COX AND DEEPSURV (CALIBRATED SURVIVAL FUNCTIONS NOT ESTIMATED IN CURRENT IMPLEMENTATION). SAGAM ACHIEVES THE BEST VALUE IN BOTH METRICS AMONG ALL STACKING APPROACHES.

Type	Model	Ctd-AUC	IBS
Base	RSF	0.634 $\pm$ 0.068	0.181 $\pm$ 0.028
	GBS	0.614 $\pm$ 0.069	0.197 $\pm$ 0.042
	XGBoost	0.626 $\pm$ 0.067	—
	DeepSurv	0.650 $\pm$ 0.062	—
Stack	Linear	0.642 $\pm$ 0.066	0.184 $\pm$ 0.030
	SAGAM	0.650 $\pm$ 0.054	0.177 $\pm$ 0.023

TABLE V
PER-FOLD TEST C-INDICES (5-FOLD NESTED CV, TCGA-LUAD). AMONG BASE LEARNERS AND STACKING METHODS, SAGAM IS THE TOP-PERFORMING MODEL IN FOUR OF FIVE FOLDS. STAGE-ONLY COX (NOT SHOWN) IS STRONGEST OVERALL (0.657 $\pm$ 0.044).

Fold	RSF	GBS	XGB	DS	Linear	SAGAM
1	0.602	0.545	0.554	0.593	0.609	0.639
2	0.584	0.584	0.620	0.592	0.584	0.604
3	0.625	0.589	0.539	0.680	0.642	0.605
4	0.706	0.718	0.646	0.716	0.716	0.718
5	0.581	0.590	0.654	0.599	0.586	0.603
Mean	0.619	0.605	0.602	0.636	0.627	0.634

and ≈ 36 test events per fold, the study is powered to detect C-index differences of ≈ 0.11 ; the observed $\Delta = +0.006$ lies below this threshold, indicating that the observed internal difference is too small to distinguish reliably from noise in the current sample.

Table IV reports time-dependent concordance AUC (Ctd-AUC), a discrimination metric robust to censoring, computed at the 25th, 50th, and 75th percentiles of training event times per fold, alongside Integrated Brier Score (IBS), a time-dependent prediction-error metric requiring survival function estimates. **SAGAM ties DeepSurv for the highest mean Ctd-AUC (0.650) and achieves the best Ctd-AUC among stacking-based approaches**, with the lowest fold-to-fold variance (std 0.054) among all stacking methods, mirroring its C-index performance pattern and providing independent support that SAGAM improves discrimination over linear stacking. **SAGAM also achieves the lowest IBS (0.177 $\pm$ 0.023), indicating lower time-dependent prediction error compared to linear stacking (0.184 $\pm$ 0.030) and RSF (0.181 $\pm$ 0.028).** IBS was not reported for XGBoost-Cox and DeepSurv because the current implementation did not estimate calibrated survival functions for these models using a common baseline hazard procedure; future work will evaluate survival-function calibration for all base learners under a unified baseline hazard estimation framework.

TABLE VI

FULL 4-LEARNER SAGAM SPLINE DF SENSITIVITY. META-LEARNER REFITTED ACROSS DF VALUES USING *fixed* OOF BASE LEARNER PREDICTIONS FROM THE MAIN PIPELINE; THESE VALUES ARE NOT DIRECTLY COMPARABLE TO TABLE III (0.634), WHICH RERUNS THE FULL NESTED TRAINING PROCEDURE PER FOLD. DF = 4 ACHIEVES THE BEST MEAN C-INDEX AND LOWEST FOLD-TO-FOLD VARIANCE WITHIN THIS ABLATION.

Variant	Mean C	Std
SAGAM df=3	0.610	0.061
SAGAM df=4	0.620	0.038
SAGAM df=5	0.612	0.052
SAGAM df=6	0.609	0.045

D. Spline Degree-of-Freedom Ablation

Table VI shows SAGAM performance across spline complexity settings, evaluated on the full four-learner ensemble by refitting only the meta-learner with each df value using fixed OOF base learner predictions from the main pipeline. df=4 achieves the best mean C-index and lowest variance; higher df values show increasing fold-to-fold variance consistent with overfitting when the spline basis becomes too flexible relative to ≈ 145 training events.

df=3 also underperforms, indicating insufficient flexibility to capture the non-monotone smooth functions visible in Fig. 3.

E. Clinical Risk Stratification

Fig. 2 shows Kaplan-Meier curves for pooled SAGAM OOF risk tertiles across all 501 TCGA patients. The low-risk group demonstrates markedly superior survival versus the high-risk group (log-rank $p = 6.94 \times 10^{-5}$; multivariate $p = 2.52 \times 10^{-4}$). These pooled OOF predictions were never used during any training step, confirming generalization to held-out patients.

F. Feature Block Analysis

The Cox baseline results (Table III) reveal a counterintuitive pattern: adding features to a Cox model monotonically *reduces* discrimination. Stage-only Cox (0.657) > clinical-only (0.646) > clinical+genomic (0.641) = all-features (0.641). This consistent ordering suggests that added features do not improve the penalized Cox baseline in this setting.

We attribute this pattern to two mechanisms. First, *feature collinearity*: several genomic and hypoxia features are correlated with AJCC stage (e.g., aneuploidy increases with advancing stage; hypoxia scores correlate with tumor size). In a penalized linear model, collinear features dilute the stage coefficient rather than providing additive predictive information. Second, *regularization masking*: CoxNet with an ℓ_1 penalty shrinks many added features to zero, yet even weak partial correlations among retained features introduce estimation variance that outweighs their marginal predictive contribution.

SAGAM’s pipeline addresses a different problem by routing all features through nonlinear base learners (RSF, GBS,

TABLE VII

NONLINEARITY OF SAGAM SMOOTH FUNCTIONS f_k (MEAN ACROSS 5 FOLDS). $\text{NONLINFRAC} = 1 - R_{\text{LINEAR}}^2$; RMSD = DEVIATION FROM BEST-FIT LINE. ALL FOUR BASE LEARNERS SHOW SUBSTANTIAL NONLINEARITY, CONFIRMING THAT A FIXED SCALAR WEIGHT CANNOT ADEQUATELY REPRESENT THEIR CONTRIBUTION.

Base Learner	NonlinFrac	RMSD
RSF	0.349	0.208
GBS	0.402	0.153
XGBoost	0.337	0.254
DeepSurv	0.424	0.276

XGBoost, DeepSurv) before meta-learning. Each base learner can identify nonlinear interactions between stage, genomic, and hypoxia features that are invisible to a Cox model with linear main effects. The GAM meta-learner then integrates these nonlinear representations through smooth functions. This architecture explains why Table IX shows consistent selection of Aneuploidy Score, Buffa Hypoxia, and TMB—features that add no value to a linear Cox model. Importantly, SAGAM does not exceed the strongest Cox baselines in internal C-index; its contribution is instead improved nonlinear stacking over scalar-weighted ensembles and interpretable model-level contribution functions unavailable from any Cox model.

These results suggest that molecular and transcriptomic features may contain nonlinear prognostic structure not captured by linear Cox models. However, because SAGAM does not improve discrimination beyond AJCC stage, this signal should be interpreted as useful for model-level analysis and nonlinear ensemble behavior rather than demonstrated incremental clinical value. SAGAM suggests that nonlinear base learners may extract structure from these features, but the current results do not demonstrate incremental discrimination beyond AJCC stage.

G. Smooth Contribution Analysis

Fig. 3 shows the smooth functions $f_k(s)$ from outer fold 1. All four functions are nonlinear, validating the core SAGAM assumption that scalar weights inadequately capture the meta-learning integration.

DeepSurv and RSF exhibit the steepest gradients, consistent with their highest individual C-indices. GBS and XGBoost show flatter curves, indicating partial redundancy with stronger learners. The non-monotone shape for DeepSurv—saturating at high risk scores—is a score-range-dependent pattern that a fixed positive scalar weight cannot express, justifying the B-spline parameterization.

H. Nonlinearity Quantification

Table VII quantifies the nonlinearity of each smooth function f_k across all five outer folds. For each fold, we evaluate $f_k(s)$ on a 300-point grid, fit a linear regression to $(s, f_k(s))$, and compute (i) the nonlinearity fraction $1 - R_{\text{linear}}^2$ (fraction of f_k variance not explained by a linear fit) and (ii) the root mean squared deviation (RMSD) from the linear fit.

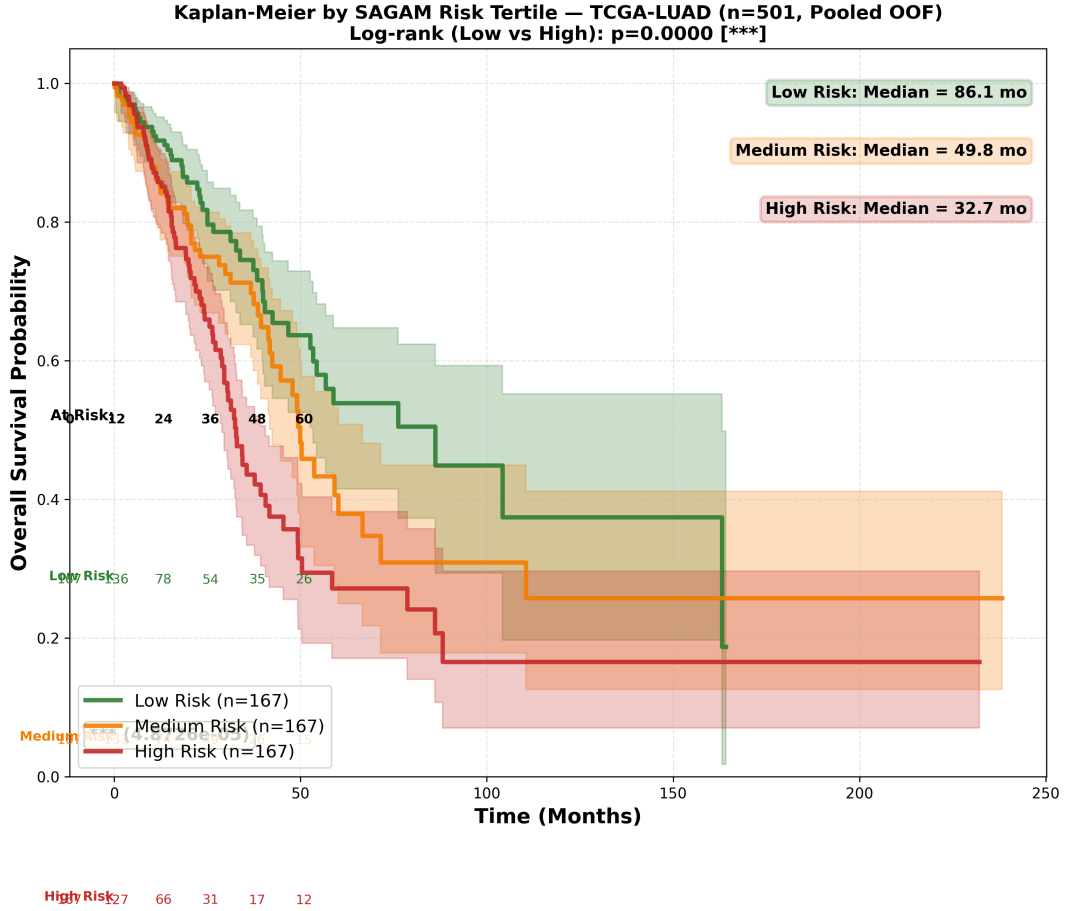


Fig. 2. Kaplan-Meier overall survival by SAGAM risk tertile with number-at-risk table. Pooled OOF predictions from 5-fold nested CV, TCGA-LUAD (n=501, 181 events). Log-rank (low vs. high): $p = 6.94 \times 10^{-5}$ (***).

All four base learners exhibit substantial nonlinearity (Non-linFrac > 0), with DeepSurv showing the largest deviation consistent with its non-monotone curve in Fig. 3. These results confirm that a fixed scalar weight γ_k —equivalent to a linear function $f_k(s) = \gamma_k s$ —inadequately represents the base learner’s contribution across the full risk spectrum.

Fig. 4 shows the smooth functions f_k across all five outer folds. The nonlinear shapes are consistent across folds for all four base learners, confirming that the observed nonlinearity reflects stable structure in the data rather than fold-specific artifacts.

Fig. 5 shows the time-dependent Brier score (outer fold 1), confirming that SAGAM’s lower time-dependent prediction error over linear stacking and RSF is consistent across the follow-up horizon.

I. Subgroup Analysis by AJCC Stage

Table VIII reports pooled OOF C-indices stratified by AJCC pathological stage using all 501 TCGA patients. SAGAM’s advantage over linear stacking is largest in Stage I/II patients

($\Delta = +0.010$, n=394, 120 events), where molecular and genomic features may contribute to heterogeneous base learner behavior, and smaller in Stage III/IV ($\Delta = +0.005$, n=105, 61 events), where advanced-stage clinical signals dominate and linear combination already captures most available information. This stage-dependent pattern is consistent with the GSE31210 (stage I/II only) external result ($\Delta = +0.086$) and the GSE68465 (all stages) result ($\Delta = -0.026$): SAGAM’s nonlinear integration provides the most benefit when base learner performance is heterogeneous.

J. Feature Selection Stability

Table IX reports the ten most stably selected features. AJCC stage, Aneuploidy Score, and Buffa Hypoxia Score are selected in all five folds. Interestingly, the Cox baseline analysis (Table III) shows that adding genomic and hypoxia features to a Cox model does not improve over stage-only Cox, yet these features are consistently selected as part of the full SAGAM feature matrix. This suggests that RSF, GBS, XGBoost, and DeepSurv can extract nonlinear interactions

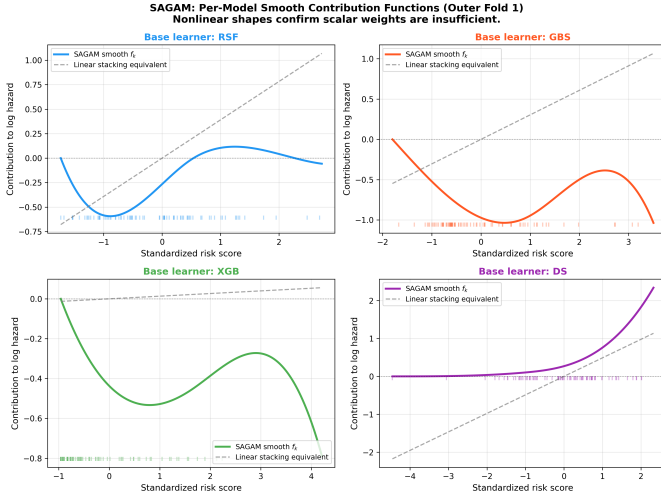


Fig. 3. SAGAM smooth contribution functions $f_k(s)$ per base learner (outer fold 1), each in its own panel. Standardized x-axis; dashed line shows linear stacking equivalent. All four functions are nonlinear, validating the B-spline parameterization.

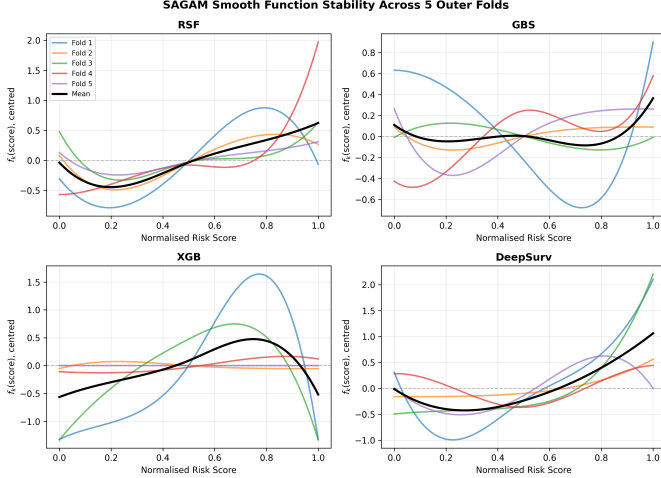


Fig. 4. SAGAM smooth function f_k stability across all 5 outer folds. Each panel shows one base learner; thin lines = individual folds; thick black line = mean. The consistent nonlinear shapes across folds confirm that the nonlinearity is a stable data property, not a fold-specific artifact.

among these features that the linear Cox model cannot, and SAGAM’s smooth meta-learning then integrates those signals.

K. External Validation on Two Independent Cohorts

Table X reports concordance indices on two independent GEO cohorts, with SAGAM trained exclusively on TCGA. GSE31210 (Takeuchi et al., 226 stage I/II pure LUAD, 35 OS events) and GSE68465 (Director’s Challenge Consortium, 432 mixed-stage LUAD, 226 OS events) represent qualitatively different external test settings.

On GSE31210, SAGAM (0.596) outperforms linear stacking by +0.086—a larger observed advantage than the internal +0.006—with significant risk stratification ($p=0.030$; Fig. 6). On GSE68465, SAGAM (0.549) does not outperform linear

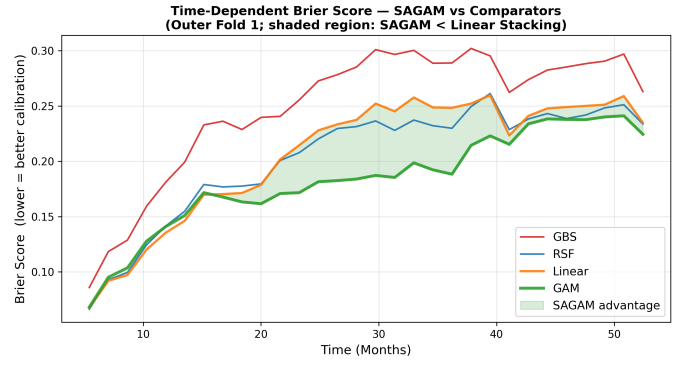


Fig. 5. Time-dependent Brier score (outer fold 1; lower = lower time-dependent prediction error). Shaded region: time points where SAGAM < Linear Stacking. Mean IBS across all 5 folds: SAGAM 0.177, Linear 0.184, RSF 0.181, GBS 0.197 (Table IV).

TABLE VIII
SUBGROUP ANALYSIS BY AJCC STAGE (POOLED OOF, 501 TCGA PATIENTS). SAGAM’S ADVANTAGE IS LARGEST IN STAGE I/II, WHERE SAGAM SHOWS ITS LARGEST OBSERVED ADVANTAGE OVER LINEAR STACKING.

Group	n	Events	C(GAM)	C(Lin)
Stage I/II	394	120	0.574	0.564
Stage III/IV	105	61	0.582	0.577
All patients	501	181	0.625	0.619

stacking (0.575, $\Delta = -0.026$, NS). This cohort-level divergence is clinically interpretable: in GSE68465, all four base learners perform similarly (RSF 0.576, DS 0.572, GBS 0.554, XGB 0.548), leaving little room for score-range-dependent reweighting; in GSE31210, DeepSurv (0.626) dominates other base learners (GBS 0.477), and SAGAM’s smooth functions can effectively up-weight the dominant learner while down-weighting the weaker ones. This pattern is also consistent with the internal subgroup analysis (Section IV-I): SAGAM’s advantage is largest in Stage I/II ($\Delta = +0.010$, where base learner performance is most heterogeneous) compared to Stage III/IV ($\Delta = +0.005$).

V. DISCUSSION

AJCC stage is the strongest predictor, internally and externally. Stage-only Cox dominates all models in both evaluations: 0.657 internally and 0.678 externally on GSE31210, exceeding DeepSurv (0.626) and SAGAM (0.596) on the external cohort. Combining SAGAM risk scores with stage (Stage + SAGAM = 0.653) does not improve over stage alone (0.657), confirming that SAGAM’s learned risk scores do not carry substantial incremental prognostic information beyond AJCC staging. These findings reinforce that SAGAM’s contribution is methodological, not clinical superiority: it provides an interpretable nonlinear stacking mechanism that improves over linear stacking and exposes per-model contribution functions, but should not be presented as a replacement for AJCC staging in clinical decision-making.

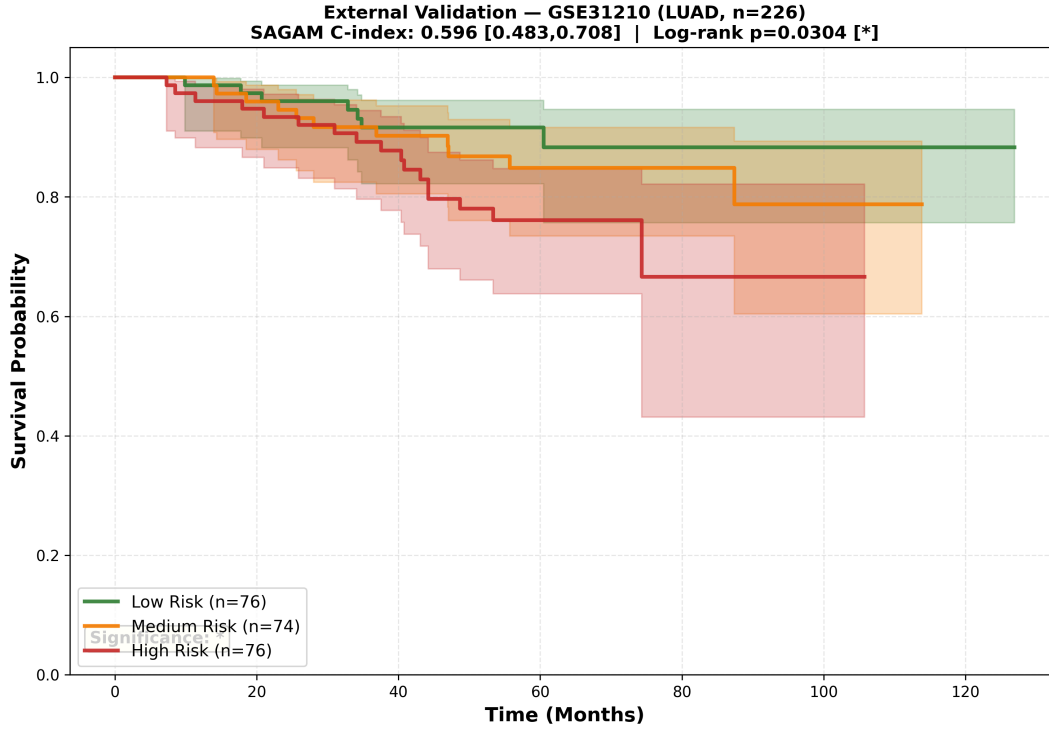


Fig. 6. External validation Kaplan-Meier curves on GSE31210 (n=226, stage I/II pure LUAD, 35 OS events). SAGAM trained on TCGA only. Log-rank (low vs. high): $p = 0.030$ (*).

TABLE IX
TOP-10 STABLY SELECTED FEATURES (5-FOLD NESTED CV)

Feature	Folds	Category
AJCC Pathologic Tumor Stage	5/5	Clinical
Aneuploidy Score	5/5	Genomic
Buffa Hypoxia Score	5/5	Hypoxia
Age at Diagnosis	5/5	Clinical
TMB (nonsynonymous)	4/5	Genomic
MSI MANTIS Score	4/5	Genomic
Ragnum Hypoxia Score	4/5	Hypoxia
Path T Stage	4/5	Clinical
Tumor Break Load	3/5	Genomic
Path N Stage	3/5	Clinical

What SAGAM adds over linear stacking. All prior survival stacking methods [8]–[11], [13] assign fixed scalar weights to base learner predictions. SAGAM replaces scalar weights with smooth B-spline functions, allowing the meta-learner to capture score-range-dependent integration rules (1). **The strongest supportive evidence comes from GSE31210, where SAGAM outperforms linear stacking by +0.086, compared with +0.006 internally; GSE68465 shows this benefit depends on cohort structure and base-learner heterogeneity.** When base-learner score distributions shift between training and test cohorts, linear stacking’s fixed scalar weights may become miscalibrated across the full risk spectrum. SAGAM’s smooth functions may adapt the integration rule to different score ranges, but the mixed external results

TABLE X
EXTERNAL VALIDATION ON TWO INDEPENDENT LUAD COHORTS. SAGAM OUTPERFORMS LINEAR STACKING ON STAGE I/II LUAD (GSE31210, $\Delta = +0.086$) BUT NOT ON THE ALL-STAGE DIRECTOR’S CHALLENGE COHORT (GSE68465, $\Delta = -0.026$), CONSISTENT WITH STAGE-DEPENDENT SUBGROUP PATTERNS OBSERVED INTERNALLY. [†]KM LOG-RANK p TESTS SAGAM RISK-GROUP SEPARATION, NOT C-INDEX SUPERIORITY OVER LINEAR STACKING. BOOTSTRAP CIs REPORTED FOR STACKING MODELS; BASE-LEARNER CIs WERE OMITTED TO FOCUS UNCERTAINTY REPORTING ON THE PRIMARY STACKING COMPARISON (FUTURE WORK WILL REPORT UNCERTAINTY FOR ALL EXTERNAL MODELS). STAGE-ONLY COX NOT EVALUATED ON GSE68465 (STAGE NOT SUFFICIENTLY HARMONIZED).

Model	GSE31210		GSE68465	
	Stage I/II, $n = 226$, 35 ev.		All stages, $n = 432$, 226 ev.	
	C	CI	C	CI
Stage-only Cox	0.678	—	stage not harmonized	
RSF	0.541	—	0.576	—
GBS	0.477	—	0.554	—
XGBoost-Cox	0.538	—	0.548	—
DeepSurv	0.626	—	0.572	—
Linear Stacking	0.510	[0.390, 0.616]	0.575	[0.533, 0.612]
SAGAM	0.596	[0.483, 0.709]	0.549	[0.508, 0.589]
Δ C-index	+0.086		−0.026	
KM log-rank $p^{\dagger}$	0.030		0.153	

indicate that this benefit depends on cohort composition and base-learner heterogeneity. SAGAM also achieves the lowest Integrated Brier Score (IBS=0.177) among all models with

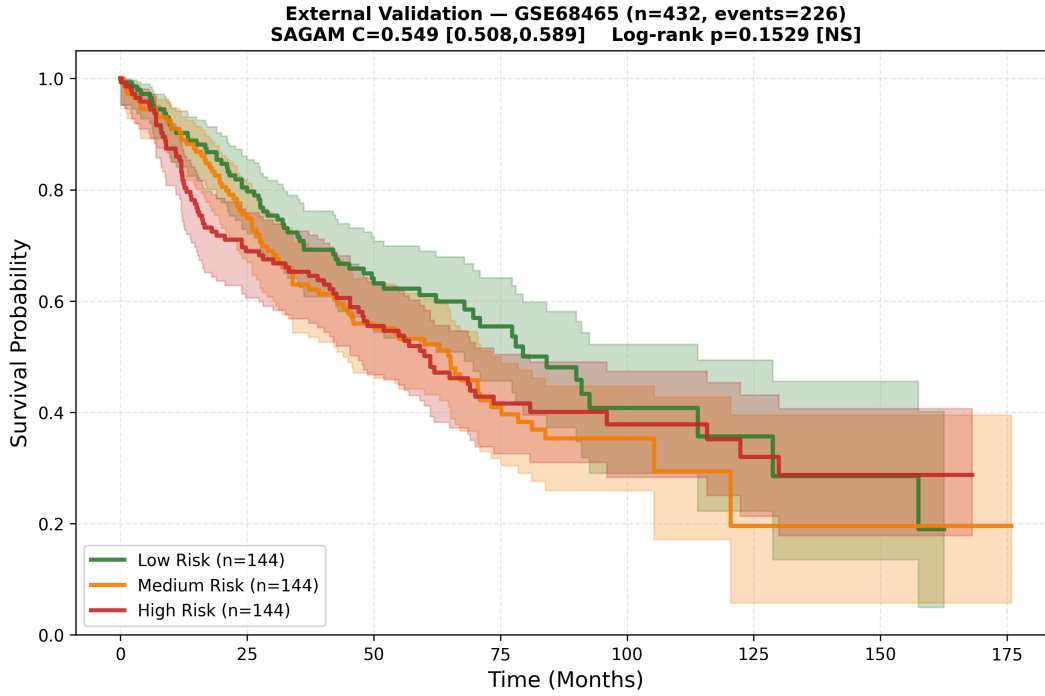


Fig. 7. External validation Kaplan-Meier curves on GSE68465 (Director’s Challenge Consortium; n=432, all stages, 226 OS events). SAGAM trained on TCGA only ($\Delta(\text{SAGAM} - \text{Linear}) = -0.026$, NS). Unlike the stage I/II GSE31210 cohort, base learners perform within a narrow band (0.548–0.576), leaving little room for nonlinear reweighting.

TABLE XI
DEEPSURV VS. SAGAM: COMPLEMENTARY ROLES

Criterion	DeepSurv	SAGAM
Internal mean C-index	0.636	0.634
GSE31210 C-index	0.626	0.596
GSE68465 C-index	0.572	0.549
Fold-to-fold std	0.058	0.050
Ensemble integration	no	yes (4 learners)
Interpretable contribution	no	smooth f_k
Advantage over linear stk	n/a	+0.006 (int); +0.086 (GSE31210)

comparable IBS support, indicating lower time-dependent prediction error compared to linear stacking (0.184) and RSF (0.181), providing further support that SAGAM’s integration reduces prediction error beyond C-index alone. The smooth functions (Fig. 3) confirm that all four base learners exhibit nonlinear relationships to the ensemble log-hazard, providing qualitative validation of the B-spline parameterization (see also Section IV-G for quantitative nonlinearity analysis).

DeepSurv as the strongest base learner. DeepSurv achieves the highest raw C-index externally (0.626) and ties SAGAM internally (0.636 vs. 0.634). Table XI contrasts SAGAM and DeepSurv along multiple dimensions. SAGAM’s primary value is not raw C-index superiority but interpretable nonlinear ensemble integration: its smooth contribution functions reveal where in the risk spectrum each model contributes, information unavailable from a neural black-box.

Stage-dependent external advantage. External validation on two independent cohorts reveals a nuanced picture. On GSE31210 (stage I/II, 35 events), SAGAM outperforms linear stacking by +0.086—a larger observed advantage than the internal +0.006. On GSE68465 (Director’s Challenge, all stages, 226 events), SAGAM does *not* outperform linear stacking ($\Delta = -0.026$, NS). This divergence aligns with the internal subgroup analysis: SAGAM’s advantage is +0.010 in Stage I/II (where base learner performance is most heterogeneous) versus +0.005 in Stage III/IV (where clinical stage dominates and linear combination already captures most available information). The mechanism is further supported by base learner heterogeneity: on GSE31210, DeepSurv (0.626) substantially outperforms GBS (0.477), creating a heterogeneous performance landscape that SAGAM’s smooth functions can exploit; on GSE68465, all learners perform within a narrow band (0.548–0.576), leaving little gain for nonlinear reweighting. These results establish a clearer scope for SAGAM: *its nonlinear integration is most beneficial when base learners have heterogeneous performance profiles.*

Feature interactions and nonlinear signal. A key finding from Table III is that linear Cox models fail to benefit from molecular features (genomic, hypoxia) beyond what stage alone provides. Yet Table IX shows these same features are stably selected across SAGAM’s folds. This apparent contradiction resolves when one recognizes that RSF and GBS can extract nonlinear interactions among stage, Aneuploidy Score, hypoxia scores, and TMB that are invisible to a linear

model. SAGAM's smooth meta-learning then integrates these nonlinearly-extracted signals, explaining why it substantially outperforms linear stacking even with a modest internal C-index gain.

Scalability and meta-learner complexity. SAGAM's meta-feature matrix has 16 columns (4 spline bases $\times$ 4 learners), compared to 4 for linear stacking. Meta-learner fitting is fast: CoxNet on 16 features with ≈ 400 training observations completes in under one second. The marginal cost over linear stacking is negligible, while the gain in interpretability and potential performance under covariate shift is substantial. This favorable complexity-benefit tradeoff supports SAGAM as a practical alternative to linear stacking when interpretability and score-range-dependent integration are desired.

Limitations. First, SAGAM's internal improvement over linear stacking is directionally consistent but not statistically significant (Wilcoxon $p = 0.313$; bootstrap CI $[-0.013, +0.025]$), reflecting limited power with 181 OS events. Second, AJCC stage-only Cox achieves the highest C-index both internally (0.657) and externally (0.678), and combining SAGAM risk scores with stage does not improve over stage alone (0.653 vs. 0.657). SAGAM therefore contributes an interpretable nonlinear stacking mechanism, not incremental clinical value beyond staging. Future work should investigate whether richer feature sets or larger cohorts yield incremental discrimination. Third, DeepSurv remains the strongest base learner by raw C-index in both evaluations, and in fold 3 it substantially outperforms the SAGAM ensemble. Fourth, external validation results are stage-dependent: SAGAM outperforms linear stacking on GSE31210 (35 OS events, stage I/II, $\Delta = +0.086$, $p=0.030$) but not on GSE68465 (226 OS events, all stages, $\Delta = -0.026$, NS), establishing the scope of SAGAM's generalization. The GSE31210 estimate also carries wide uncertainty (bootstrap CI width 0.226) due to limited events. Fifth, the spline df sensitivity was evaluated with the full four-learner ensemble (Table VI), supporting $df=4$ as the best-performing setting in this sensitivity analysis; a cross-validated sensitivity for individual learner contributions remains future work. Sixth, SAGAM models additive smooth effects without cross-model interaction terms; tensor-product splines and adaptive ensemble pruning are natural extensions. Seventh, the external results should not be interpreted as universal evidence of improved transportability: SAGAM improves over linear stacking on GSE31210 but underperforms it on GSE68465, and the scope of generalization depends on cohort composition and base-learner performance heterogeneity.

VI. CONCLUSION

SAGAM introduces interpretable nonlinear survival stacking by replacing scalar ensemble weights with smooth B-spline functions of base learner risk scores within a Cox hazard model. To the best of our knowledge, SAGAM is the first survival stacking framework to use GAM-style smooth functions of base learner risk scores as the Cox meta-learning mechanism. On TCGA-LUAD ($n=501$, 5-fold nested CV), SAGAM (0.634) achieves the best C-index, lowest IBS

(0.177), and highest Ctd-AUC (0.650) among all stacking-based approaches, outperforms linear Cox stacking in four of five folds, and provides lower fold-to-fold variance than all individual learners; AJCC stage-only Cox (0.657) remains the strongest internal baseline. Overall, SAGAM provides an interpretable nonlinear alternative to scalar-weighted survival stacking. It improves over linear stacking internally across C-index, Ctd-AUC, and IBS, and shows a larger observed advantage on the stage I/II GSE31210 cohort, but not on the all-stage GSE68465 cohort. These mixed external results indicate that SAGAM's benefit is cohort-dependent and most likely when base learners exhibit heterogeneous performance. Because AJCC stage-only Cox remains the strongest discriminator and Stage + SAGAM does not improve over stage alone, SAGAM should be viewed as a methodological and interpretability contribution rather than an incremental clinical predictor beyond staging.

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